1.1 Part A: Whitening and autocovariance matrices

1.1.1 Conventions and implementation

The observations are stored as the rows of an $n \times p$ matrix X. Each variable is centred using its full-sample column mean,

$$X_c = X - \mathbf{1}_n \bar{x}^{\mathsf{T}}, \qquad S_0 = \frac{1}{n-1} X_c^{\mathsf{T}} X_c.$$

If $S_0 = VDV^{\mathsf{T}}$, the symmetric whitening matrix and whitened data are

$$W = S_0^{-1/2} = VD^{-1/2}V^{\mathsf{T}}, \qquad Y = X_c W.$$

Eigenvalues below 10^{-10} times the largest eigenvalue are treated as numerically zero, because their inverse square roots would be unstable. No additional marginal scaling is applied: scaling is already included in W.

For lag $h \geq 1$, the implementation uses

$$\widehat{C}_h = \frac{1}{n-h} \sum_{t=1}^{n-h} y_t y_{t+h}^{\mathsf{T}}.$$

The mean is not re-estimated inside each lag window. The default matrix used for subsequent orthogonal diagonalisation is the symmetric version $\widehat{C}_h^{(s)} = (\widehat{C}_h + \widehat{C}_h^{\mathsf{T}})/2$. Thus lagging pairs row t with row $t+h$, and normalization uses the number of available pairs, $n-h$.

1.1.2 Purpose of whitening

Whitening removes contemporaneous scale and cross-covariance, so the transformed sample has lag-zero covariance I_p. In the second-order source-separation model this changes the unknown mixing matrix into an orthogonal one. The remaining directions can therefore be estimated by orthogonal diagonalisation of the lagged autocovariance matrices. Whitening does *not* turn the series into white noise: it removes dependence between variables at lag zero, while serial dependence at non-zero lags remains in $\widehat{C}_h$ and supplies the signal information.

1.1.3 Numerical verification

For the supplied data, $n = 1000$ and $p = 10$. The numerical check gave

$$\max_j |\bar{y}_j| = 6.49 \times 10^{-14}, \qquad \max_{j,k} |\{\widehat{\mathrm{Cov}}(Y) - I_{10}\}_{jk}| = 3.55 \times 10^{-15}.$$

The whitened variances ranged from 0.999999999999997 to 1, and the largest absolute off-diagonal covariance was 2.58×10^{-15}. These values confirm the expected effect to numerical precision. The non-zero Frobenius norms of the lagged matrices (for example, 0.3653 at lag 1 and 0.5575 at lag 2) also show that whitening does not force non-zero-lag covariance estimates to zero.

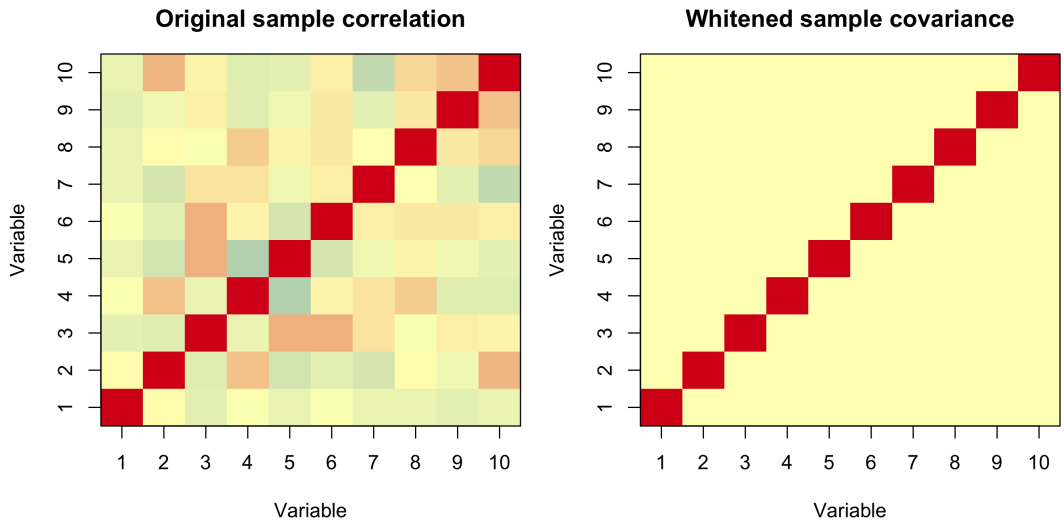


Figure 1: Original contemporaneous correlation and covariance after whitening.

1.2 Part B: Bootstrap for dependent observations

1.2.1 Fixed-block bootstrap

For block length L, a block start is sampled uniformly with replacement from $\{1, \ldots, n - L + 1\}$. The corresponding L consecutive multivariate rows are retained in their original order. This is repeated $\lceil n/L \rceil$ times, the blocks are concatenated, and the result is truncated to n rows. The tuning parameter L controls the amount of local dependence retained within a block.

1.2.2 Stationary bootstrap

The first index is uniform on $\{1, \ldots, n\}$. At each later time, a new uniform index is selected with probability $r = 1/L$; otherwise the preceding index is advanced by one, with circular wrap-around from n to 1. Hence the block length K is geometric,

$$\Pr(K = k) = r(1 - r)^{k-1}, \qquad E(K) = \frac{1}{r} = L.$$

Unlike the fixed-block method, block lengths are random and circular continuation avoids a special boundary. In 100 generated samples, target expected lengths 10, 20 and 40 produced mean observed lengths 9.94, 20.20 and 40.94, respectively.

1.2.3 Why iid resampling is unsuitable

Sampling individual rows independently destroys their temporal ordering and therefore makes non-zero-lag autocovariances approximately zero. The resulting bootstrap sample no longer represents the statistic's variability under the original dependent process. Block methods resample consecutive multivariate observations and retain short-range serial and cross-series dependence within blocks.

1.2.4 Effect of block length

Dependence preservation was measured by the mean absolute difference between the original and bootstrap sample autocorrelations, averaged over all 10 variables and lags 1–20. Each entry below

summarizes 100 bootstrap samples.

Table 1: Mean absolute ACF error; smaller values indicate better local-dependence preservation.

Method	Mean error	Standard deviation
iid	0.0460	0.0017
Fixed, $L = 10$	0.0381	0.0021
Fixed, $L = 20$	0.0320	0.0019
Fixed, $L = 40$	0.0270	0.0025
Stationary, $E(K) = 10$	0.0336	0.0020
Stationary, $E(K) = 20$	0.0295	0.0024
Stationary, $E(K) = 40$	0.0269	0.0027

The iid method has the largest error. Longer blocks preserve the observed ACF more closely, although they also yield fewer effectively independent blocks and slightly greater Monte Carlo variability. The expected block length is therefore a bias–variance tuning choice, not a parameter that should automatically be made as large as possible.

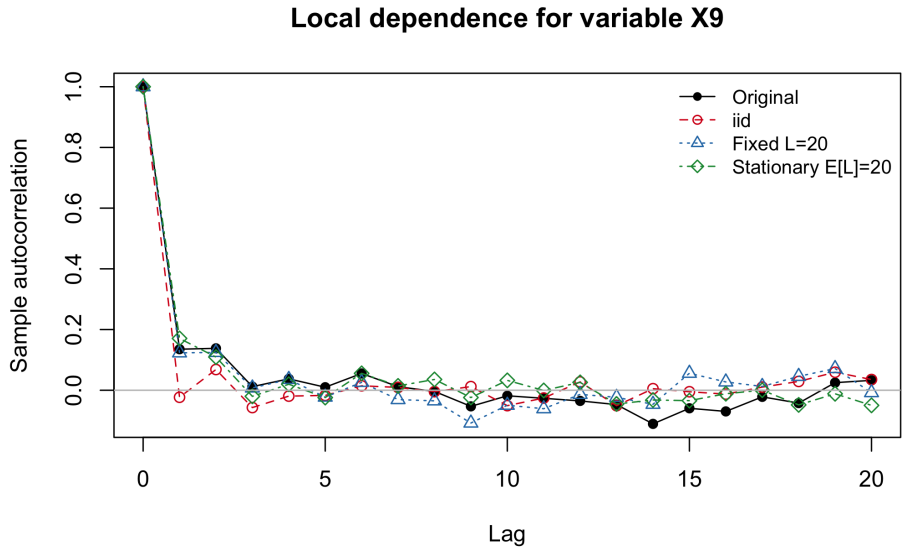


Figure 2: Example ACF comparison for the variable with the largest absolute lag-1 sample ACF.

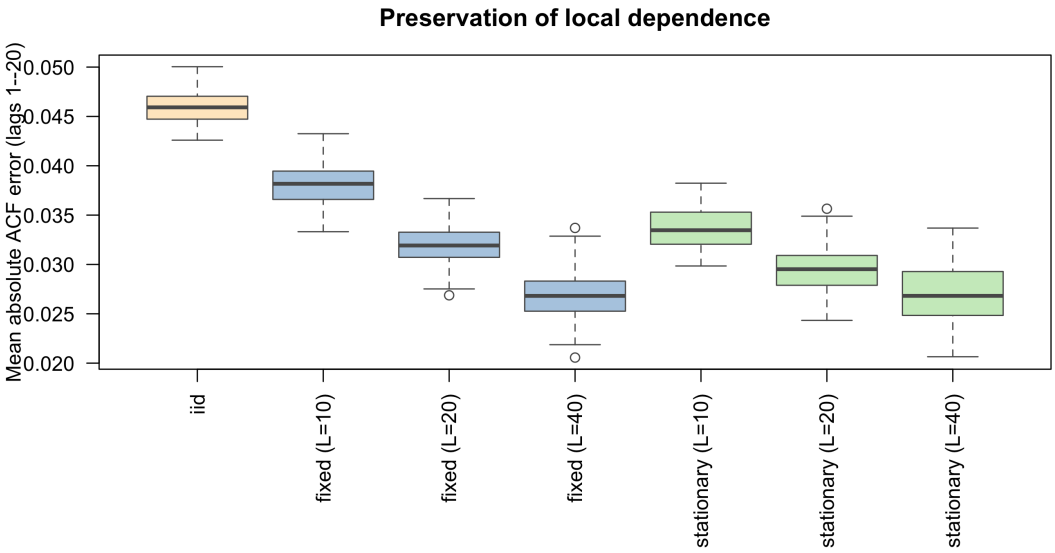


Figure 3: ACF error over 100 bootstrap samples for each method and block length.

1.3 Part C: The ladle quantities

1.3.1 Eigenvalue-based quantity ϕ

For the lag set $\mathcal{T}$, an orthogonal matrix U approximately jointly diagonalises the whitened autocovariance matrices. The component strengths are

$$s_j = \sum_{h \in \mathcal{T}} \left[\operatorname{diag}\left(U^{\mathsf{T}} \widehat{C}_h^{(s)} U \right)_j \right]^2,$$

ordered so that $s_0 \geq \cdots \geq s_{p-1}$. The normalized scree contribution is

$$\phi(k) = \frac{s_k}{1 + \sum_{\ell=0}^{p-1} s_\ell}, \qquad k = 0, \ldots, p-1.$$

Large values represent directions with serial dependence; noise directions should have strengths close to zero. With several lags, U is obtained using `JADE::rjd`; the strength calculation and ordering are implemented directly.

1.3.2 Bootstrap-variability quantity f

Let $\widehat{B}_k$ and $B^*_{k,b}$ contain the first k ordered directions from the original and bth bootstrap samples. The subspace instability is

$$f_0(k) = \frac{1}{B} \sum_{b=1}^{B} \left\{ 1 - \left| \det\left(\widehat{B}_k^{\mathsf{T}} B^*_{k,b} \right) \right| \right\}, \quad k = 1, \ldots, p-1, \qquad f_0(0) = 0,$$

and

$$f(k) = \frac{f_0(k)}{1 + \sum_{\ell=0}^{p-1} f_0(\ell)}.$$

The determinant measures agreement of the two k-dimensional subspaces and is unaffected by sign changes of individual directions. Stable signal subspaces have small $f(k)$; instability increases after noise directions enter the subspace.

1.3.3 Combination and dimension estimate

The ladle criterion and estimator are

$$g(k) = f(k) + \phi(k), \qquad \widehat{q} = \underset{k \in \{0,\ldots,p-1\}}{\arg\min}\; g(k).$$

Thus ϕ decreases when weak or noise directions are included, while f increases when the estimated subspace becomes unstable. Their sum balances these two pieces of information.

1.3.4 Simulation check

The code was checked using Setting 1 of Nordhausen and Virta (2018): five Gaussian AR/MA signal series and five Gaussian white-noise series, so $p = 10$ and the true dimension is $q = 5$. The check used $n = 1000$, lags 1:12, 100 bootstrap replications, and block length 20. Both bootstrap versions returned $\widehat{q} = 5$. The largest whitening identity error was 3.22×10^{-15}, the largest orthogonality error in $U^{\mathsf{T}}U$ was 2.22×10^{-15}, and the joint off-diagonal squared-energy ratio was 0.0718. These checks establish correct whitening, orthogonal directions, useful approximate joint diagonalisation, and recovery of the known signal dimension.

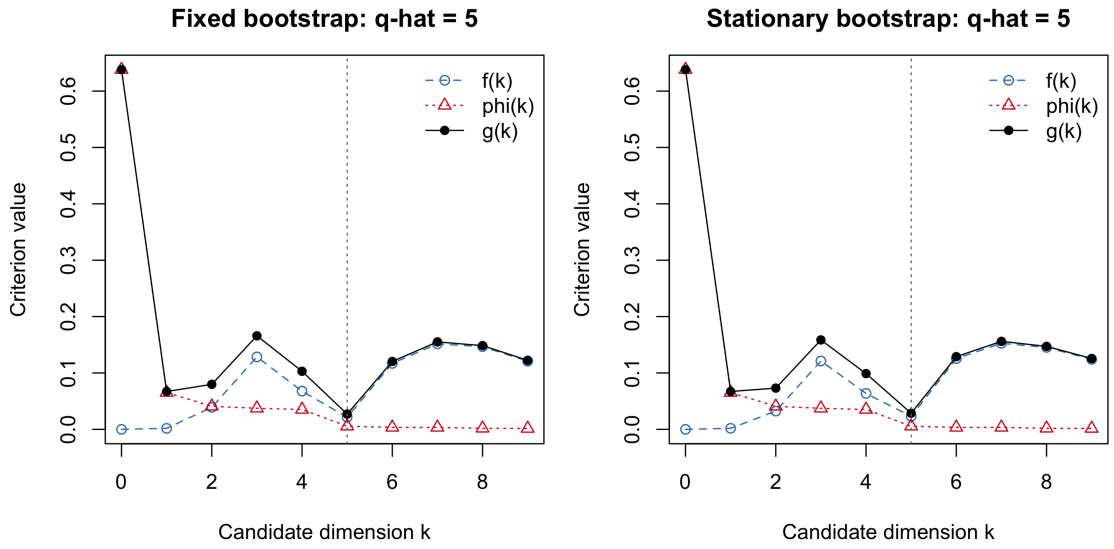


Figure 4: Implemented f, ϕ and g on simulated Setting 1 data. The minimum is at the true dimension $q = 5$ for both bootstrap methods.